AP PRECALCULUS

Unit 1 • Lesson 1.10 notes

Lesson 1.10 • Rational Functions and Holes

Lesson notes • Learning goals, key ideas, worked examples, common mistakes, quick check

Learning goals

- Identify holes in the graph of a rational function.
- Find the coordinates of a hole and the limit of the function there.

Vocabulary. hole, removable discontinuity, common factor, simplified form.

1. Holes come from cancellation

Key idea

If $(x-a)$ is a factor of the numerator at least as many times as of the denominator, the graph has a **hole** at $x=a$. Its height is the value of the simplified expression at a, which also equals $\lim_{x\to a} r(x)$.

If the factor appears *more* times in the denominator, it does not fully cancel and $x=a$ is a vertical asymptote instead.

Example 1

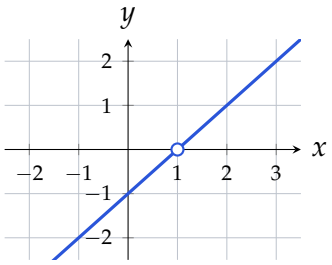
$$r(x)=\frac{x^2+2x-8}{x^2-4}=\frac{(x+4)(x-2)}{(x+2)(x-2)}.$$

Solution. Hole at $x=2$, height $\frac{6}{4}=\frac{3}{2}$: the hole is $\left(2,\frac{3}{2}\right)$, and $\lim_{x\to 2} r(x)=\frac{3}{2}$. Vertical asymptote $x=-2$. Zero $x=-4$.

Example 2 • A hole on the x-axis

$$r(x)=\frac{(x-1)^2}{x-1}.$$

Solution. $r(x)=x-1$ for $x\neq 1$. The hole is at $(1,0)$. Although the simplified expression is 0 at $x=1$, r has *no* zero there, because $r(1)$ is undefined.



Common mistakes

- Finding the hole's y-coordinate by substituting into the original (unsimplified) expression, which gives $\frac{0}{0}$.
- Calling a hole a zero.

Quick check

1. Where is the hole of $\dfrac{x^2 - x - 6}{x - 3}$? 2. Does $\dfrac{x - 4}{(x - 4)^2}$ have a hole at $x = 4$?

Answers. 1. $(3, 5)$. 2. No: $\frac{1}{x-4}$ remains, so $x = 4$ is a vertical asymptote.